A.U.R.A.

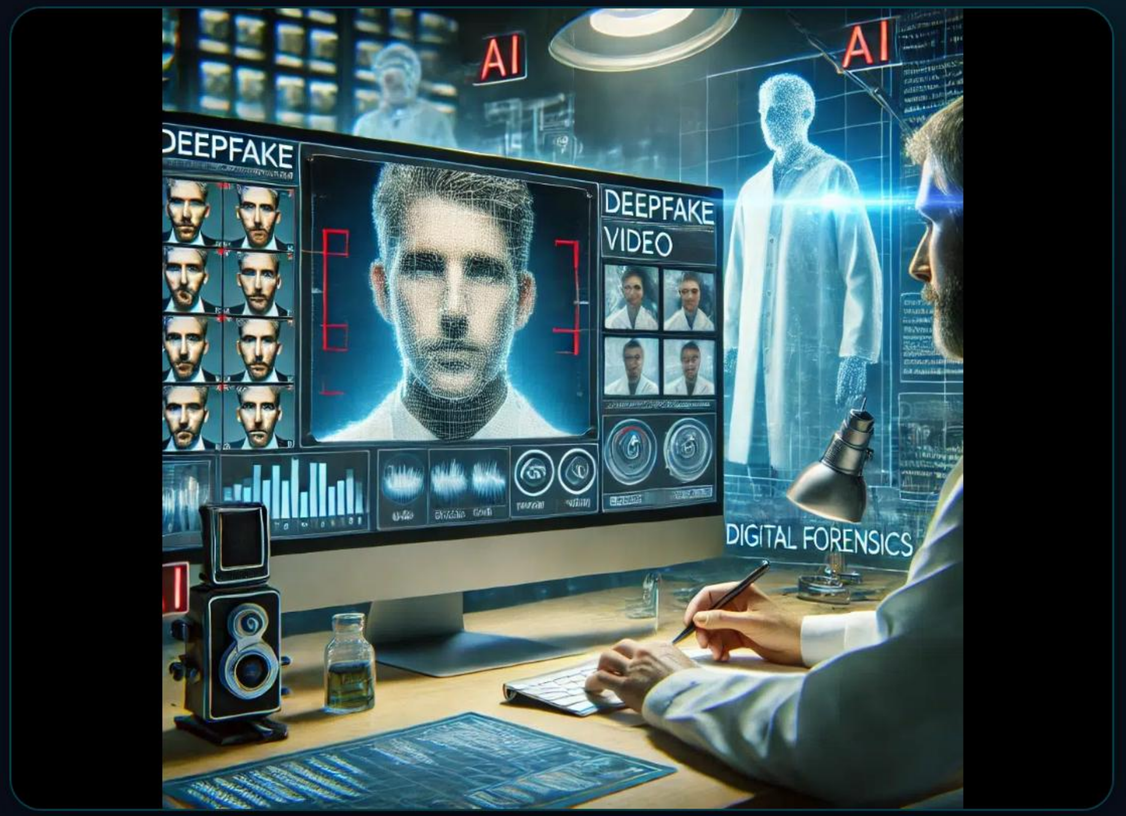
Authenticated Unforgeable Reality Architecture

The **"ISI Mark"** for Digital Reality.



Generative AI Has Destroyed Visual Truth

- **The Illusion of Reality:** Deepfakes and AI image generators are now indistinguishable from actual photographs.
- **Fragile Metadata:** Standard metadata (EXIF) embedded in images can be stripped or manipulated by bad actors in seconds.
- **The Massive Impact:** This inability to verify visual truth has led to unprecedented spikes in insurance fraud, fake civic grievance reports, and completely bypassed KYC (identity) verification systems.



Why Current Defenses Are Failing



The Hardware Trap (Big Tech)

Major tech companies are attempting to solve this by installing expensive "secure silicon" chips directly into phone cameras to cryptographically sign photos at the hardware level.

The Flaw: This approach completely excludes 80% of the global market that relies on budget smartphones without specialized security hardware.



The Software Trap (Current Apps)

Current verification apps rely solely on basic Operating System data (like GPS coordinates or timestamps) to validate media.

The Flaw: Highly vulnerable to "Rooted Device" attacks. Hackers can easily inject fake GPS data and pipe deepfake video directly into the camera feed, bypassing OS-level checks entirely.

Our Innovation: Environmental Cryptography

- **Hardware Agnostic:** We completely abandon the need for expensive, specialized hardware chips.
- **Utilizing Existing Tech:** We harness the standard MEMS sensors already built into every smartphone—even budget Android devices.
- **Capturing Physics:** A.U.R.A. records the exact Light (Lux), Magnetism (μT), and 3D Gravity of the environment at the exact millisecond the shutter opens.
- **The Ultimate Truth:** You can fake pixels using AI, but **you cannot fake the laws of physics.**



The A.U.R.A. Pipeline

1

Secure Capture

Extracts raw RGB pixel data alongside live environmental sensor telemetry simultaneously.

2

On-Device ML Engine

AI instantly analyzes the collected sensor data to detect any physical anomalies or inconsistencies.

3

Cryptographic Sealing

The pixel data and Sensor JSON are mathematically fused via an unbreakable SHA-256 hash.

4

Watermarking

The hash is embedded into invisible metadata and stamped onto the image as a verifiable visual QR code.

Beating the Screen Attack

"What if a hacker simply takes a picture of a deepfake displayed on a high-res laptop monitor?"

Our ML Defense Mechanism:

- **Photometer Analysis:** Instantly detects the invisible 240Hz PWM (Pulse-Width Modulation) flicker emitted by laptop and monitor screens.
- **Magnetometer Defense:** Detects the heavy, unnatural electromagnetic interference generated by the monitor's internal circuitry.
- **Visual Model:** Identifies Moiré pixel grid patterns unique to capturing digital screens.

Result: The capture is instantly rejected.



Who Needs A.U.R.A.?

Insurance & Auto

Enabling zero-fraud collision and property damage claims by ensuring all submitted photographic evidence is physically verified at the moment of capture.

Smart City / Civic Tech

Guaranteeing 100% verified pothole, infrastructure damage, and civic grievance reporting to optimize municipal resource allocation.

Fintech & KYC

Providing unspoofable live-selfie identity verification to prevent sophisticated presentation attacks during financial onboarding.

Journalism

Offering cryptographic proof for field reporters operating in conflict zones, ensuring visual evidence cannot be discredited as AI manipulation.

Engineered for the Edge

Frontend & Sensors

Built on React Native / Flutter with low-level Native SensorManager hooks for zero-latency extraction of raw environmental telemetry.

Machine Learning

Powered by TensorFlow Lite for on-device, real-time anomaly detection, ensuring privacy and eliminating server round-trip latency.

Backend & Crypto

Robust Python/FastAPI architecture handling secure SHA-256 Hashing, cryptographic key management, and EXIF metadata injection.

Storage & Ledger

Utilizing PostgreSQL paired with a Web3 Ledger (Polygon/Solana) to generate and store permanent, immutable public receipts of authenticity.

THE FUTURE STANDARD OF TRUTH

Thank You. Q&A.

Before the ISI mark, helmet safety was a guess.

A.U.R.A. is the ISI mark for digital media.

Hardware-Agnostic | Democratized Security | Mathematically Unforgeable

Image Sources



https://img.freepik.com/premium-photo/glowing-red-digital-lock-dark-smoky-backdrop-cybersecurity-online-safety-protection-3d-rendering_117466-54250.jpg?semt=ais_hybrid&w=740&q=80

Source: www.freepik.com



https://cybercentaurs.com/wp-content/uploads/Deepfake.webp

Source: cybercentaurs.com



https://cdn2.botland.store/129181-large_default/dfrobot-gravity-line-tracking-sensor-reflective-digital.jpg

Source: botland.store



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Source: www.istockphoto.com